

EduGenie-Google Gemini Powered Learning Assistant

Government Arts and Science College-Manur-MSU142

Team Leader : Gayathri E

Team Members: Uma Maheswari S, Reka M, Nithersana M, Abinaya M

Requirement Analysis Phase

The requirement analysis phase is the foundation of the Software Development Life Cycle (SDLC). It is the process of defining exactly what a new system or software needs to do, translating vague business goals into precise technical blueprints before any code is written.

If this phase is rushed, it inevitably leads to scope creep, budget overruns, and building features no one actually needs.

The Four Core Stages

Requirement analysis is a strictly sequential process. You cannot document a requirement until you have analysed it, and you cannot analyse it until you have gathered it.

1.Elicitation (Gathering):Collecting needs from all stakeholders.

This involves figuring out what the client, end-users, and business sponsors actually want. Common techniques include:

- **Interviews and workshops:** Direct conversations with stakeholders.
- **Shadowing:** Observing users in their current workflow to see where bottlenecks exist.
- **Questionnaires:** Gathering broad feedback from a large user base.

2. Analysis and Negotiation: Resolving conflicts and feasibility.

Stakeholders often ask for contradictory features or things that violate budget or timeline constraints. In this step, business analysts:

- Identify and resolve conflicting requests.
- Categorize requirements into "Must-haves," "Should-haves," and "Nice-to-haves."
- Conduct technical and financial feasibility checks.

3. Specification (Documentation): Drafting the blueprint.

The finalized, agreed-upon needs are translated into formal documentation. This usually results in a **Software Requirements Specification (SRS)** document. The SRS outlines both:

- **Functional requirements:** What the system must *do* (e.g., "The system must process payments via Stripe").
- **Non-functional requirements:** How the system must *perform* (e.g., "The system must load pages in under 2 seconds").

4. Validation: Securing final sign-off.

Before the design and development teams take over, the SRS document is presented back to the stakeholders. They review the specifications to confirm that the documented plan perfectly aligns with their original vision and business needs. Once approved, it becomes the official baseline for the project.